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BY

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Autonomous Visual Docking of a BlueROV2: Simulation Terminal Guidance and Empirical Limits of ArUco Perception Underwater

Thesis submitted as a requirement for the degree of Bachelor of Engineering

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Abstract

This thesis develops the optical terminal stage of a vision-based underwater docking system for a low-cost ROV. A perception–guidance–control pipeline using position-based visual servoing is shown in simulation to dock a BlueROV2 autonomously onto a swaying dock through coarse-approach and fine-alignment phases. In parallel, the operating envelope of passive ArUco perception is characterised from real underwater imagery: detection range, pose error, and degradation with distance and turbidity. Together these bound the realistic operating range of marker-based terminal docking and identify where a complementary far-field cue becomes necessary.

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